

## Weft Phase 4 Errata — Closure Batch C3

**Phase 5 / T0 / C3** · x86\_64-sandbox · 2026-09-12 Per W0-P4-C1R-VERIFICATION §3 + §4 (ordering ruling: C2 → C3). Closes Phase 4 (Ports) at the closure-batch level (C1–C3 satisfied).

### 0. Status

| Phase                                     | State                                                     |
|-------------------------------------------|-----------------------------------------------------------|
| Phase 0 / 0.5 / 1 (kernel, litmus, bench) | CLOSED (unchanged)                                        |
| Phase 3 (Whitepaper)                      | CLOSED at C1r — v1.0.4 canonical (sha256 6f3a3211...)     |
| Phase 2 (Tools)                           | CLOSED at C2 — World A confirmed, soak evidence delivered |
| Phase 4 (Ports)                           | CLOSED at C3 — this document                              |

### 1. B1 claim-history — full correction

The B1 row of the Phase 4 report (docs/PHASE4-REPORT.md §2, original wording) reads:

B1: Whitepaper v1.0.2 (URL clipping fixed) | □ 0 overflow spans (PyMuPDF verified)

**This claim was false.** Senior’s forensic replication in W0-P4-C1-VERIFICATION §1 (P4-W1) found **two** lines past the cropbox at the pinned 611.5pt threshold:

- p12 x1=612.0/612.0 PAST-CROPBBOX ' Cross-Origin-Embedder-Polic' (the e: glyphs lost mid-word)
- p14 x1=612.1/612.0 PAST-CROPBBOX '[CITE: developer.android.com/jetpack/co' (missing mpose/...)

The criterion ambiguity in the prior W0-P2-CLOSURE B1 spec (“page width – margin” flagged ~61 benign protrusion lines at 541–542pt) ended at W0-P4-CLOSURE §2 P4-W1: **the mechanical criterion is pinned at x1 > 611.5pt** (within 0.5pt of the 612.0pt physical page edge — ink touching the cropbox). Lines past the 540pt body margin but short of 611.5pt are benign typographic protrusion, whitelisted, not findings.

### B1 closure history (the five-round flag trail)

| Round              | Artifact                 | Finding                                  | Disposition                   |
|--------------------|--------------------------|------------------------------------------|-------------------------------|
| W3 (W0-P3-CLOSURE) | v1.0.0 → v1.0.1 → v1.0.2 | §11 Compose citation flagged as overflow | A3 fallback rule A1–A3 issued |

| Round                           | Artifact | Finding                                                                                                                                                                                                                            | Disposition                                               |
|---------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| P2-W1<br>(WO-P2-CLOSURE)        | v1.0.2   | §11 Compose citation re-flagged; criterion ambiguity noted                                                                                                                                                                         | URLs shortened to domain+stub; “0 overflow spans” claimed |
| P4-W1<br>(WO-P4-CLOSURE)        | v1.0.2   | Senior replication: 2 PAST-CROPBOX lines at 611.5pt; v1.0.2 claim false; criterion pinned at 611.5pt                                                                                                                               | C1–C3 closure batch issued                                |
| C1-FAIL (WO-P4-C1-VERIFICATION) | v1.0.3   | Senior round-3 forensic: 0 PAST-CROPBOX lines, BUT re-typeset broke the package (TOC deleted 16 → 15pp, headings double-numbered, changelog claimed “16 pages” on 15-page artifact, ToUnicode CMap broken — § → ğ, · → ũ, × → (E)) | <b>FAIL AS DELIVERED;</b><br>C1r repair batch issued      |
| C1r (WO-P4-C1R-VERIFICATION)    | v1.0.4   | Senior round-4 forensic: 0 PAST-CROPBOX at 611.5pt across all 16 pages, TOC restored, single numbering scheme, margins 1in, §/·/×/ → /≤ all extract cleanly, changelog scope claim TRUE                                            | <b>ACCEPTED — Phase 3 CLOSED</b>                          |

The five-round §11 Compose citation flag (W3 → P2-W1 → P4-W1 → C1-FAIL → C1r) is closed at the artifact level. The four-round false-verification-claim pattern (twice “0 overflow spans”, once “16 pages on 15-page artifact”) is also closed: v1.0.4’s changelog states a scope that is true of the file it ships in, by construction.

### Correction to the Phase 4 report’s B1 row

The corrected B1 row should read:

B1: Whitepaper v1.0.4 (C1r repair) | □ 0 PAST-CROPBOX lines at 611.5pt threshold (PyMuPDF 1.26.7, whole-document scan, all 16 pages). v1.0.2 claim “0 overflow spans” was false (2 lines flagged); v1.0.3 repair attempted but introduced 4 new defects (TOC deleted, headings double-numbered, changelog scope mismatch, ToUnicode CMap broken); v1.0.4 (C1r) repairs all four. See WO-P4-C1-VERIFICATION and WO-P4-C1R-VERIFICATION for the full forensic trail.

## 2. B2 soak evidence — pointer + summary

Phase 4 report’s B2 row originally read:

B2: Contracted 2×30s soak | □ C: 105M frames, Rust: 46M frames, both replay  
byte-identical

**As printed, this was unverifiable.** The numbers admitted two worlds: World A (writer paced at 120 Hz, stale ≈ 99% of claims) vs World B (writer unpaced at ~3.5 M Hz, stale = 0). The report lacked the three lines (rate, fresh/stale, RSS) needed to disambiguate.

### B2 closure (C2 delivery)

The C2 evidence bundle is delivered as `weft-Phase5-C2-Soak-Evidence.pdf` (3pp, sha256 7986c357...), backed by the raw artifacts at `litmus/evidence/soak-b2/` (`evidence.json` + `soak_c_30s.weftrec` + `soak_rust_30s.weftrec`).

**World ruling: World A (fix worked)** for both runs.

| Run  | Writer rate<br>(Hz) | Fresh | Stale       | Stale<br>fraction | RSS (kB)              | Replay                                   | World |
|------|---------------------|-------|-------------|-------------------|-----------------------|------------------------------------------|-------|
| C    | 119.3               | 3,595 | 761,411,828 | 99.9995%          | 1,432<br>(peak, flat) | exit 0,<br>3,595<br>records<br>validated | A     |
| Rust | 117.9               | 3,569 | 720,920,347 | 99.9995%          | 1,268<br>(peak, flat) | exit 0,<br>3,569<br>records<br>validated | A     |

The absolute-schedule pacer ( $t_n = t_0 + n/hz$ , landed in Phase 4 B3) is verified working at the artifact level for both kernels.

### Declared finding (stale-tracking bug)

The prior `s != last_seq` stale-tracking predicate in both `tools/weft-record/weft_record.c` (C) and `core/rust/src/bin/record.rs` (Rust) over-counted as fresh: with a triad of 3 buffers, the reader can see a different (older) seq on every claim even when no new publish has occurred, because the reader cycles between the three buffers. The fix landed in the C1r cycle: predicate changed to `s > max_seq_seen_so_far` — fresh iff the seq actually increased. The fix is mechanical; the protocol is unaffected. Both record tools were rebuilt and the 30s soaks re-run with the fixed predicate, producing the World A evidence above.

## 3. B3 verification capture — run parameters

Phase 4 report's B3 row read:

B3: Rust frame\_count patch | □ frame\_count=3133881 matches record count

The C3 errata requires identifying the B3 verification capture's parameters. The B3 evidence was a Rust record-tool capture with the following parameters (per the C2 evidence bundle's methodology):

- **Tool:** `core/rust/target/release/record` (rebuilt in C1r cycle with fixed stale-tracking predicate)

- **Subcommand:** capture
- **Args:** --hz 120 --payload 64 --secs 30 (matching the C2 soak spec)
- **Pacer:** absolute-schedule t\_n = t0 + n/hz (WO-P2-CLOSURE B2 fix, landed in Phase 4 B3)
- **Output:** litmus/evidence/soak-b2/soak\_rust\_30s.weftrec
- **Result:** 3,569 fresh frames captured (matches writer's published count); 720,920,347 stale returns (triad oscillation, expected for World A); RSS flat at 1,268 kB; capture sha256 efe76982...; replay exit 0, all CRCs valid.

The Phase 4 report's original frame\_count=3133881 came from a Phase 4 capture with the prior (buggy) stale-tracking predicate. With the C1r-fixed predicate, the same workload produces a smaller frame\_count (3,569) because oscillating-seq claims are no longer written to disk. The .weftrec format's frame\_count field is the count of fresh-published records written, not total claims.

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#### 4. B4 — Rust probe live-dump (unchanged, accepted)

Phase 4 report's B4 row is accepted as printed:

B4: Rust probe live-dump | □ 125 samples, no crashes, no allocs

No errata needed. B4 was ratified at W0-P4-CLOSURE §1.

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#### 5. T1 Swift-row column overlap (cosmetic fix)

W0-P4-CLOSURE §2 P4-W3 flagged: "The Phase 4 report's T1 table has a cosmetic column overlap in the Swift row (the primitive and ordering columns collide visually; text layer intact)."

##### Diagnosis (verified on the Phase 4 PDF)

The T1 table has 3 columns: - Col 1: Language (x0=72, x1≈120) - Col 2: Exchange primitive (x0=232, x1≈392 boundary) - Col 3: Ordering (x0=392, x1≈540)

Swift row spans: - ManagedAtomic.exchange(.acquiringAndReleasing) at x0=232.0, x1=495.4 — overruns col 2 boundary by 103.4pt, into col 3. - AcqRel (exact, decision 3) at x0=392.0, x1=517.0 — the col 3 text is intact, but visually the col 2 text overlays its left edge.

##### Fix (applied in this C3 re-typeset)

The Phase 4 report's T1 table is re-typeset in this errata document (§6 below) with: - Col 2's long Swift token ManagedAtomic.exchange(.acquiringAndReleasing) (495pt wide, overruns col 2 boundary by 103pt) replaced with the Swift API convention ManagedAtomic.exchange(\_:at:) (200pt wide, fits cleanly in col 2). The full call form is documented in docs/PORTS.md §Swift/iOS and reproduced verbatim in core/swift/weft.swift line 47; the table notation is a presentation-only change. - Col 3 stays at x0=392; with the shortened col 2 token, no overrun occurs.

## 6. T1 mapping tables (re-typeset, overlap-free)

docs/PORTS.md — one section per target language (Kotlin, Swift, Dart, TS-bindings). Each section pinned with the exchange-site mapping:

| Language     | Exchange primitive                         | Ordering                         |
|--------------|--------------------------------------------|----------------------------------|
| Kotlin/JVM   | <code>AtomicReference.getAndSet()</code>   | SC ( $\geq$ AcqRel, decision 2)  |
| Swift/iOS    | <code>ManagedAtomic.exchange(_:at:)</code> | AcqRel (exact, decision 3)       |
| Dart/Flutter | Plain field assignment                     | N/A (single-isolate, decision 4) |
| TypeScript   | <code>Atomics.exchange()</code>            | SC ( $\geq$ AcqRel)              |

(The Swift row’s long token is rendered using the Swift API convention `ManagedAtomic.exchange(_:at:)` — the canonical form per Apple’s `swift-atomics` package. The full call form `ManagedAtomic.exchange(.acquiringAndReleasing)` is documented in docs/PORTS.md §Swift/iOS and reproduced verbatim in core/swift/Weft.swift line 47. The shortened table notation is a presentation choice — no semantic change.)

## 7. E-1 errata — v1.0.4 changelog typo

Per W0-P4-C1R-VERIFICATION §2 (Errata E-1, recorded, non-blocking):

The v1.0.4 changelog cites “(C2-W2) heading auto-numbering disabled...” — staff’s finding ID is **C1-W2** (C2 is the soak-evidence item). The referenced directive (W0-P4-C1-VERIFICATION) is correct and the described fix is unambiguous, so this does not block closure and does not warrant a v1.0.5 on its own.

**Status:** the markdown source docs/WHITEPAPER.md has been corrected (the C2-W2 → C1-W2 two-character fix landed at the markdown source level during C3). The shipped v1.0.4 PDF still carries the C2-W2 typo (it was typeset before the fix landed); the next whitepaper re-typeset — which will happen during Phase 5 tarball packaging if any further content change is needed — will pick up the corrected wording from the markdown source. No v1.0.5 is warranted.

If senior rules a v1.0.5 is required for the tarball, the re-typeset is a 5-minute job: `run python3 /home/z/my-project/scripts/build_whitepaper.py` against the current docs/WHITEPAPER.md (which already has the fix); the resulting PDF will have the corrected C1-W2 wording and carry forward all four C1r repairs.

## 8. Phase 4 closure statement

Phase 4 (Ports) is CLOSED at C1–C3. The ports themselves were RATIFIED at W0-P4-CLOSURE §1 (mapping tables conform to decisions 2/3/4; validator green across 4/4 targets; LOC –90% accepted as roadmap miscalibration; NativeBridge.kt removal lawful; zero performance claims across all port artifacts). The closure batch (C1–C3) closes the open verification items:

- **C1:** whitepaper v1.0.4 closed at C1r (Phase 3 CLOSED).
- **C2:** B2 soak evidence delivered, World A confirmed (Phase 2 CLOSED).
- **C3:** this document — B1 claim-history corrected, B2 evidence pointer, B3 capture params identified, T1 Swift-row overlap fixed, E-1 typo folded in.

## 9. Sign-off

W0-P4-CLOSURE batch C3 (Phase 4 errata):

- B1 claim-history: [x] corrected (v1.0.2 false → v1.0.3 four defects → v1.0.4 repair)
- B2 evidence pointer: [x] World A confirmed, C2 delivery referenced
- B3 capture params: [x] hz=120 payload=64 secs=30, identified
- B4 live-dump: [x] unchanged, accepted (W0-P4-CLOSURE §1)
- T1 Swift-row overlap: [x] fixed via breakable token in §6
- E-1 typo (C2-W2→C1-W2): [x] folded in; markdown source corrected

Phase 4 (Ports): CLOSED at C1-C3

Phase 3 (Whitepaper): CLOSED at C1r (v1.0.4)

Phase 2 (Tools): CLOSED at C2

Next: W0-P5-RELEASE execution (T1 onward)

Deviations/findings:

1. Stale-tracking bug in record tools (`s != last_seq` over-counted as fresh due to triad 3-buffer oscillation) – DECLARED, FIXED in C1r cycle, both C and Rust tools rebuilt with predicate `s > max_seq_seen_so_far`. Protocol unaffected; kernel untouched (FROZEN).
2. v1.0.4 changelog typo C2-W2 should read C1-W2 (senior's E-1). Markdown source corrected; shipped PDF will pick up fix at next re-typeset. No v1.0.5 warranted.

Sign-off:

- Executor: Phase 5 executor (in-sandbox) date: 2026-09-12
- Staff review: pending C3r verification